

WEEK 4 PROJECT REPORT

STOCK MARKET ANALYSIS DASHBOARD USING POWER BI

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Project Title: Stock Market Data Analysis and Interactive Dashboard

Tool Used: Microsoft Power BI Desktop

Dataset: Shopify Stock Data

INTRODUCTION

During **Week 4**, the focus of the project was on developing and enhancing the interactive Power BI dashboard.

The main work completed during this week included:

- Developing **Page 2** of the dashboard
- Creating stock price and trading-volume visualizations
- Creating a **Metric Selector**
- Adding an interactive **Date slicer**
- Creating a dedicated **Stock_Tooltip** page
- Adding KPI cards to the tooltip
- Formatting and arranging dashboard elements
- Testing the interactivity of the dashboard

The previous week's work was used as the foundation for these activities.

BRIEF REFERENCE TO PREVIOUS WORK

The required stock-market data, calculated measures, moving averages, and basic dashboard components had already been prepared in the earlier weeks.

Therefore, Week 4 concentrated mainly on **interactive dashboard development and presentation** rather than repeating the previous data-preparation steps.

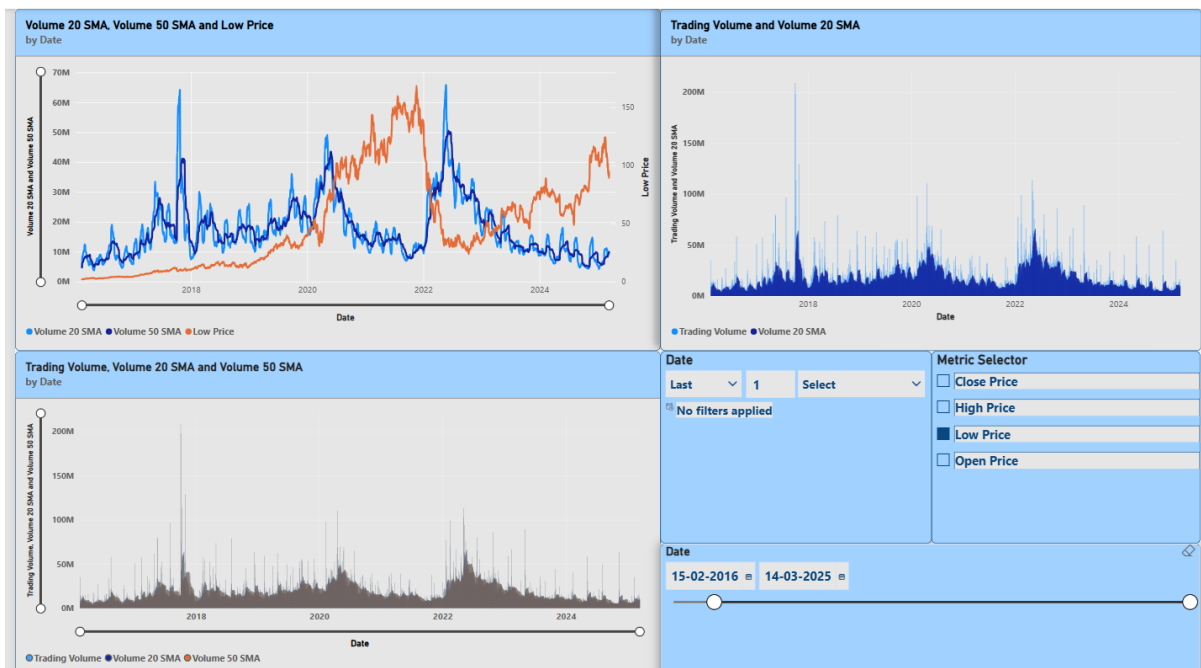
DEVELOPMENT OF PAGE 2

A second dashboard page, **Page 2** was created to provide additional analysis of stock prices and trading volume.

The page was designed with multiple charts and interactive controls.

The final Page 2 contains:

1. Volume 20 SMA, Volume 50 SMA and Low Price
2. Trading Volume and Volume 20 SMA
3. Trading Volume, Volume 20 SMA and Volume 50 SMA
4. Metric Selector
5. Date filter
6. Date-range slicer



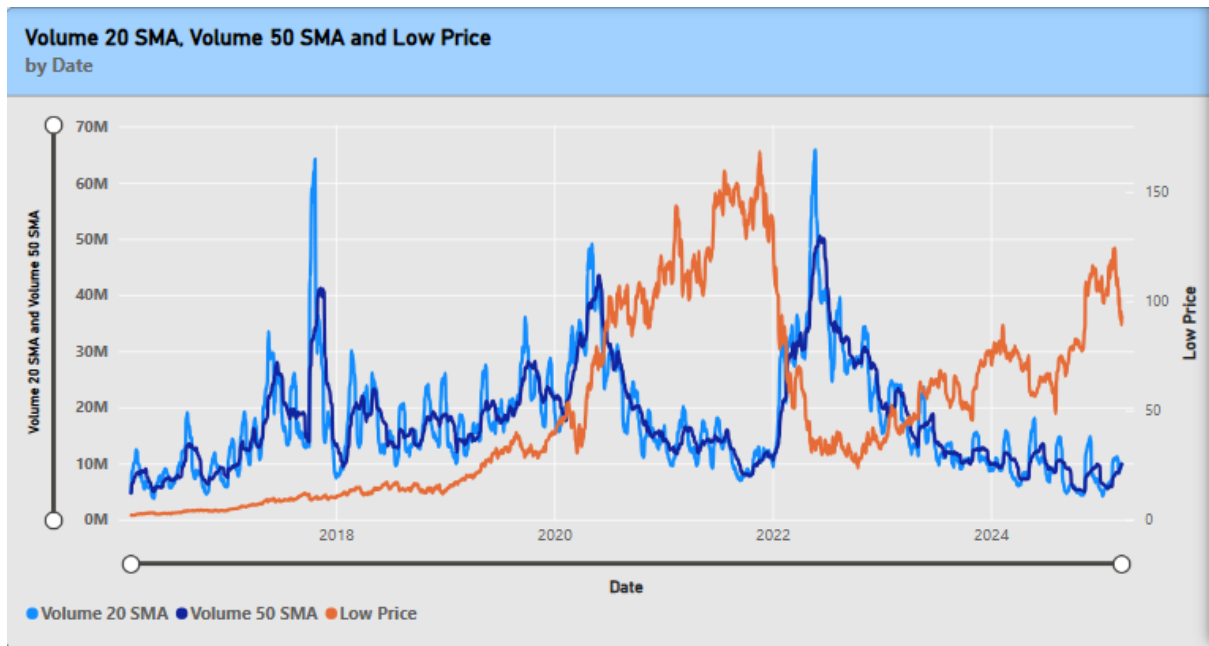
VOLUME 20 SMA, VOLUME 50 SMA AND LOW PRICE

A line chart was created using:

- **Date** on the X-axis
- **Volume 20 SMA**
- **Volume 50 SMA**
- **Low Price**

The chart provides a combined view of the stock's low price and moving-average trends over time.

The visual also uses a secondary Y-axis so that the price and volume-related measures can be displayed together.



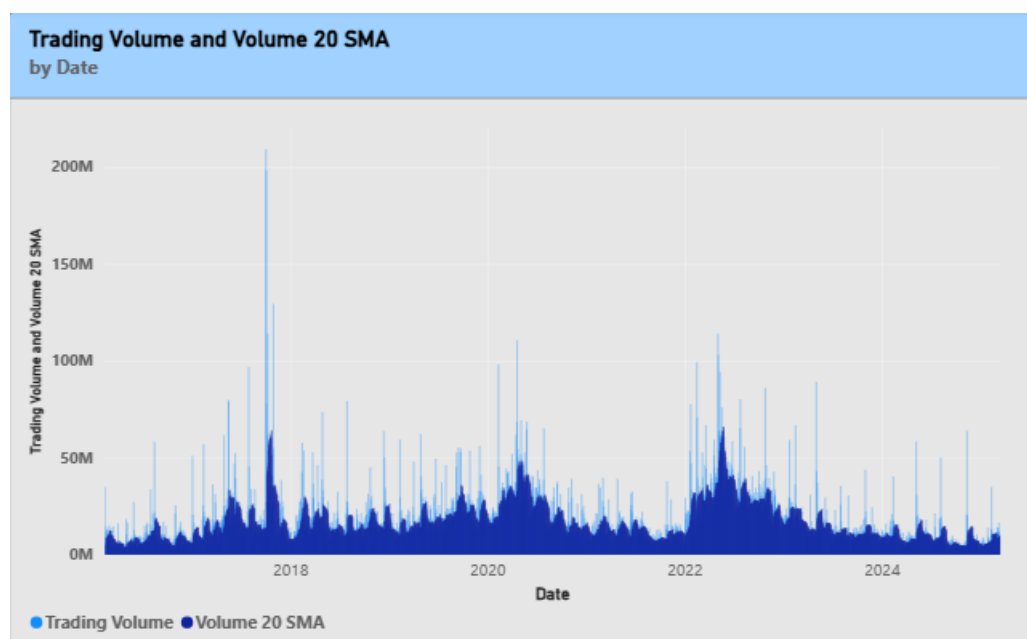
TRADING VOLUME AND VOLUME 20 SMA

A second chart was created to compare:

- Trading Volume
- Volume 20 SMA

against Date.

This visual helps compare the actual trading volume with its 20-period moving average.



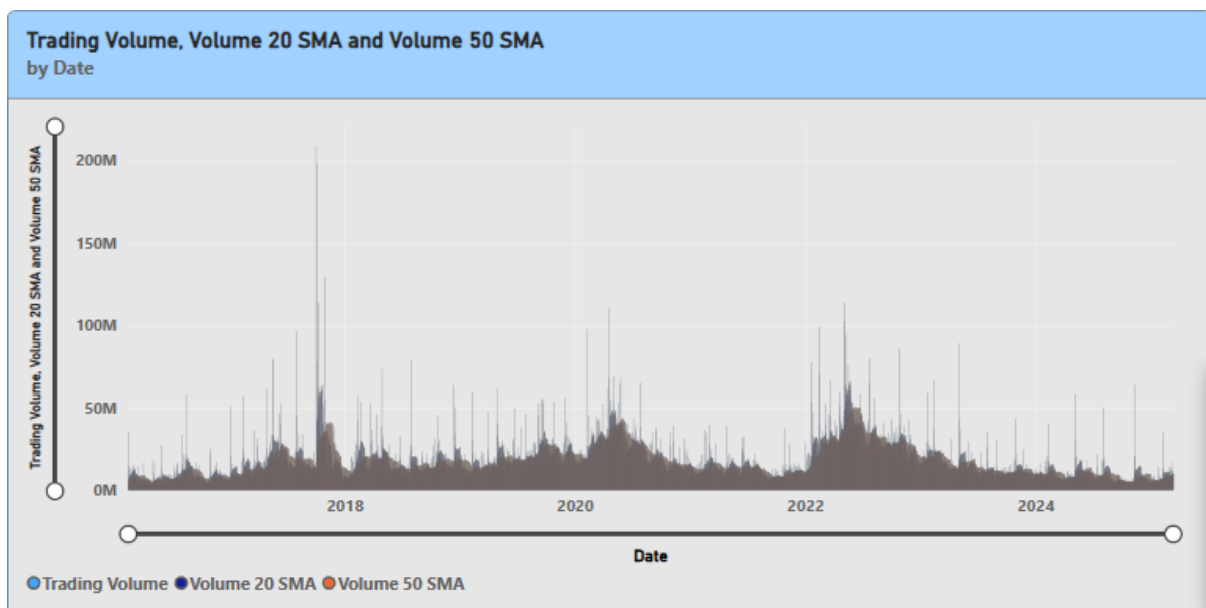
TRADING VOLUME, VOLUME 20 SMA AND VOLUME 50 SMA

A third chart was created using:

- Trading Volume
- Volume 20 SMA
- Volume 50 SMA

The chart allows the actual trading volume to be viewed together with both moving averages.

This provides a clearer view of changes and trends in trading activity.



DATE SLICER

An interactive **Date slicer** was added to Page 2.

The slicer allows the user to select a specific range of dates.

The selected date range affects the dashboard visuals, allowing users to focus on a particular period.

The slicer was configured using a date-range style with two date selection points.

The date slicer interface shows a range from 24-10-2018 to 14-03-2025. The timeline below the date boxes has two white circular selection points corresponding to the dates in the boxes.

METRIC SELECTOR

One of the main interactive features developed during Week 4 was the **Metric Selector**.

A **Field Parameter** was created in Power BI.

The parameter contains four stock-price measures:

- Close Price
- High Price
- Low Price
- Open Price

The Metric Selector allows the user to choose which price metric should be displayed in the visual.

DAX Used

```
Metric Selector = {  
    ("Close Price", NAMEOF('Measures Table'[Close Price]), 0),  
    ("High Price", NAMEOF('Measures Table'[High Price]), 1),  
    ("Low Price", NAMEOF('Measures Table'[Low Price]), 2),  
    ("Open Price", NAMEOF('Measures Table'[Open Price]), 3)  
}
```

The parameter was then added to the visual so that the chart changes according to the selected metric.



TESTING THE METRIC SELECTOR

The Metric Selector was tested by selecting different options.

The available selections were:

Close Price → High Price → Low Price → Open Price

When a metric is selected, the corresponding price information is displayed in the visual.

This made the dashboard more interactive while avoiding the need to create separate visuals for every price metric.

STOCK_TOOLTIP PAGE

A dedicated tooltip page named **Stock_Tooltip** was created.

The page was configured as a **Tooltip** page rather than a normal dashboard page.

The purpose of the tooltip is to display additional stock information when the user interacts with the relevant dashboard visual.

KPI CARDS ON STOCK_TOOLTIP

Five stock metrics were added to the tooltip:

- **Close Price**
- **High Price**
- **Low Price**
- **Open Price**
- **Trading Volume**

These were displayed using card visuals.

The cards provide a quick summary of important stock values.



TOOLTIP KPI VALUES

The completed tooltip displayed the following example values:

Metric	Displayed Value
Close Price	48.46
High Price	176.29
Low Price	1.85
Open Price	48.46
Trading Volume 41bn	

These values are displayed by the tooltip according to the dashboard's data context.

FINAL PAGE 2

The final Page 2 dashboard contains the following components:

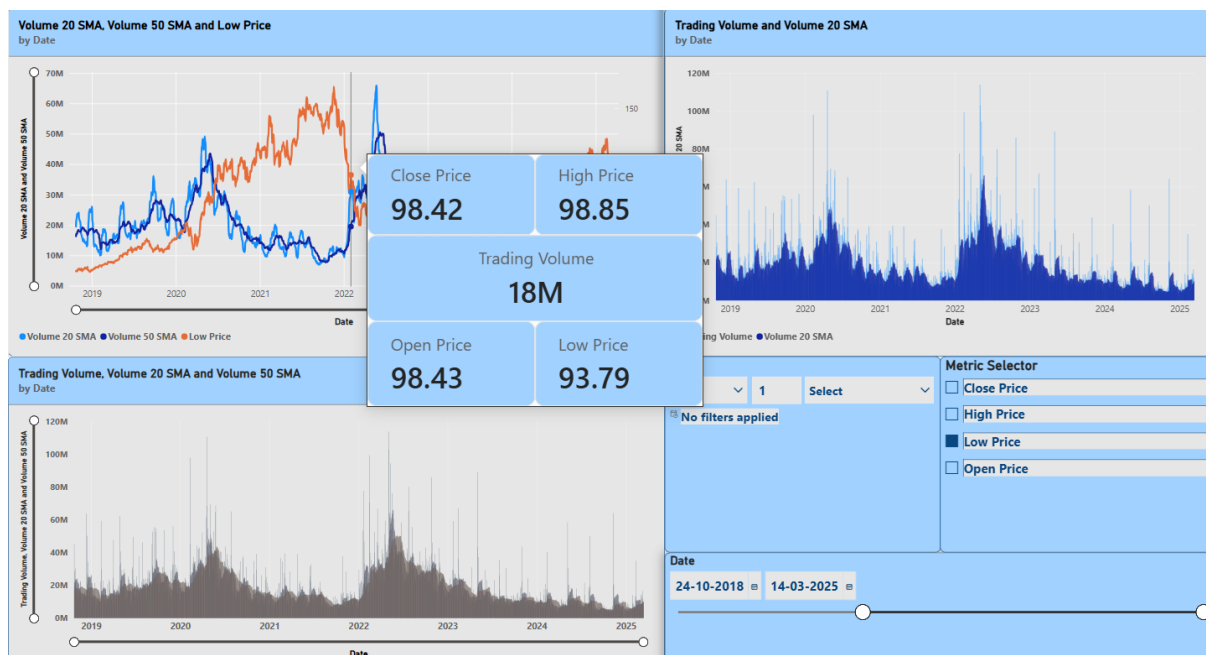
Component	Purpose
Volume 20 SMA + Volume 50 SMA + Low Price	Price and moving-average analysis
Trading Volume + Volume 20 SMA	Trading-volume trend analysis
Trading Volume + Volume 20 SMA + Volume 50 SMA Moving-average comparison	
Metric Selector	Interactive price selection
Date Slicer	Date-range filtering

FINAL STOCK_TOOLTIP

The final **Stock_Tooltip** page contains five KPI cards:

Close Price | High Price | Trading Volume | Open Price | Low Price

The tooltip provides additional information without requiring additional charts on the main dashboard.



CONCLUSION

Week 4 focused primarily on making the Power BI stock market dashboard **interactive and user-friendly**.

The major achievements were the creation of **Page 2**, the **Metric Selector**, the **Date slicer** and the **Stock_Tooltip** page with KPI cards.

The Metric Selector allows users to switch between different stock-price metrics, while the Date slicer allows users to focus on a particular period. The Stock_Tooltip provides additional stock information in a compact format.

By the end of Week 4, the dashboard had been enhanced with interactive controls, analytical charts, KPI cards, and tooltip functionality, resulting in a more complete and interactive stock-market analysis dashboard.